

Approximation

Our work in the present chapter centers around the famous theorem of Weierstrass on the approximation by polynomials of continuous real functions defined on closed intervals. This theorem, important as it is in classical analysis, has been overshadowed in recent years by a generalized form of it discovered by Stone. The latter relates to continuous functions defined on compact Hausdorff spaces, and has become an indispensable tool in topology and modern analysis.

We prove the Weierstrass theorem and then the two forms of the Stone-Weierstrass theorem which deal separately with real and complex functions. Finally, after an excursion into the theory of locally compact Hausdorff spaces, we extend the Stone-Weierstrass theorems to this context.

35. THE WEIERSTRASS APPROXIMATION THEOREM

Let us consider a closed interval $[a, b]$ on the real line and a polynomial

$$p(x) = a_0 + a_1x + \cdots + a_nx^n,$$

with real coefficients, defined on $[a, b]$.¹ Every such polynomial is obviously a continuous real function, and as a consequence of the second lemma in Sec. 20, we know that the limit of any uniformly convergent

¹ This polynomial can of course be regarded as a function defined on the entire real line. We ignore this fact and consider only x 's which lie in $[a, b]$.

sequence of such polynomials is also a continuous real function. The Weierstrass theorem states that the converse of this is also true, that is, that any continuous real function defined on $[a, b]$ is the limit of some uniformly convergent sequence of polynomials. This is clearly equivalent to the statement that such a function can be uniformly approximated by polynomials to within any given degree of accuracy. Many proofs of this classic theorem are known, and the one we give is perhaps as concise and elementary as most.

Theorem A (the Weierstrass Approximation Theorem). *Let f be a continuous real function defined on a closed interval $[a, b]$, and let $\epsilon > 0$ be given. Then there exists a polynomial p with real coefficients such that $|f(x) - p(x)| < \epsilon$ for all x in $[a, b]$.*

PROOF. As a first step, we show that it suffices to prove the theorem for the special case in which $a = 0$ and $b = 1$. If $a = b$, the conclusion follows at once on taking p to be the constant polynomial defined by $p(x) = f(a)$. We may thus assume that $a < b$. We next observe that $x = [b - a]x' + a$ gives a continuous mapping of $[0, 1]$ onto $[a, b]$, so that the function g defined by $g(x') = f([b - a]x' + a)$ is a continuous real function defined on $[0, 1]$. If our theorem is proved for the case in which $a = 0$ and $b = 1$, then there exists a polynomial p' defined on $[0, 1]$ such that $|g(x') - p'(x')| < \epsilon$ for all x' in $[0, 1]$. If we now express this inequality in terms of x , we obtain $|f(x) - p'([x - a]/[b - a])| < \epsilon$ for all x in $[a, b]$; and defining a polynomial p by $p(x) = p'([x - a]/[b - a])$ yields our theorem in the general case. Accordingly, we may assume that $a = 0$ and $b = 1$.

We next recall that if n is a positive integer and k an integer such that $0 \leq k \leq n$, then the binomial coefficient $\binom{n}{k}$ is defined by

$$\binom{n}{k} = n!/k!(n - k)!.$$

The polynomials B_n —one for each n —defined by

$$B_n(x) = \sum_{k=0}^n \binom{n}{k} x^k (1 - x)^{n-k} \left(\frac{k}{n}\right)$$

are called the *Bernstein polynomials* associated with f . We prove our theorem by finding a Bernstein polynomial with the required property.

Several identities will be needed for this. The first is a special case of the binomial theorem:

$$\sum_{k=0}^n \binom{n}{k} x^k (1 - x)^{n-k} = [x + (1 - x)]^n = 1. \quad (1)$$

If we differentiate (1) with respect to x , we get

$$\begin{aligned} \sum_{k=0}^n \binom{n}{k} [kx^{k-1}(1-x)^{n-k} - (n-k)x^k(1-x)^{n-k-1}] \\ = \sum_{k=0}^n \binom{n}{k} x^{k-1}(1-x)^{n-k-1}(k-nx) = 0, \end{aligned}$$

and multiplying through by $x(1-x)$ gives

$$\sum_{k=0}^n \binom{n}{k} x^k(1-x)^{n-k}(k-nx) = 0. \quad (2)$$

On differentiating (2) with respect to x and considering $x^k(1-x)^{n-k}$ as one of the two factors in applying the product rule, we get

$$\sum_{k=0}^n \binom{n}{k} [-nx^k(1-x)^{n-k} + x^{k-1}(1-x)^{n-k-1}(k-nx)^2] = 0. \quad (3)$$

Applying (1) to (3) gives

$$\sum_{k=0}^n \binom{n}{k} x^{k-1}(1-x)^{n-k-1}(k-nx)^2 = n;$$

and on multiplying this through by $x(1-x)$, we find that

$$\sum_{k=0}^n \binom{n}{k} x^k(1-x)^{n-k}(k-nx)^2 = nx(1-x),$$

or, on dividing both sides by n^2 ,

$$\sum_{k=0}^n \binom{n}{k} x^k(1-x)^{n-k} \left(x - \frac{k}{n}\right)^2 = \frac{x(1-x)}{n}. \quad (4)$$

Identities (1) and (4) will be our main tools in showing that $B_n(x)$ is uniformly close to $f(x)$ for all sufficiently large n .

Now for the proof of the fact just stated. By using (1), we see that

$$f(x) - B_n(x) = \sum_{k=0}^n \binom{n}{k} x^k(1-x)^{n-k} \left[f(x) - f\left(\frac{k}{n}\right) \right],$$

so that

$$|f(x) - B_n(x)| \leq \sum_{k=0}^n \binom{n}{k} x^k(1-x)^{n-k} \left| f(x) - f\left(\frac{k}{n}\right) \right|. \quad (5)$$

Since f is uniformly continuous on $[0,1]$, we can find a $\delta > 0$ such that $|x - k/n| < \delta \Rightarrow |f(x) - f(k/n)| < \epsilon/2$. We now split the sum on the

right of (5) into two parts, denoted by Σ and Σ' , where Σ is the sum of those terms for which $|x - k/n| < \delta$ (we think of x as fixed but arbitrary) and where Σ' is the sum of the remaining terms. It is easy to see that $\Sigma < \epsilon/2$. We complete the proof by showing that if n is taken sufficiently large, then Σ' can be made less than $\epsilon/2$ independently of x . Since f is bounded, there exists a positive real number K such that $|f(x)| \leq K$ for all x in $[0,1]$. From this it follows that

$$\Sigma' \leq 2K \sum \binom{n}{k} x^k (1-x)^{n-k},$$

where the sum on the right—denote it by Σ'' —is taken over all k such that $|x - k/n| \geq \delta$. It now suffices to show that if n is taken sufficiently large, then Σ'' can be made less than $\epsilon/4K$ independently of x . Identity (4) shows that

$$\delta^2 \Sigma'' \leq \frac{x(1-x)}{n},$$

so

$$\Sigma'' \leq \frac{x(1-x)}{\delta^2 n}.$$

The maximal value of $x(1-x)$ on $[0,1]$ is $1/4$, so

$$\Sigma'' \leq \frac{1}{4\delta^2 n}.$$

If we take n to be any integer greater than $K/\delta^2\epsilon$, then $\Sigma'' < \epsilon/4K$, $\Sigma' < \epsilon/2$, $|f(x) - B_n(x)| < \epsilon$ for all x in $[0,1]$, and our theorem is fully proved.

The Weierstrass theorem clearly amounts to the assertion that for any closed interval $[a,b]$ on the real line, the polynomials are dense in the metric space $\mathcal{C}[a,b]$. This is the form of the theorem which we shall generalize in the next section to $\mathcal{C}(X,R)$, where X is an arbitrary compact Hausdorff space.

The slightly restricted statement that the polynomials are dense in $\mathcal{C}[0,1]$ has another generalization, in a different direction. This result is so remarkable that we state it because of its intrinsic interest, though we give no proof. The Weierstrass theorem for $\mathcal{C}[0,1]$ says, in effect, that all real linear combinations of the functions

$$1, x, x^2, \dots, x^n, \dots$$

are dense in $\mathcal{C}[0,1]$, where by a real linear combination of these functions we mean the result of choosing any finite set of them, multiplying each by a real number, and adding. Instead of working with all positive powers of x , let us permit gaps to occur, and consider the infinite set of functions

$$1, x^{n_1}, x^{n_2}, \dots, x^{n_k}, \dots,$$

the n_k 's being positive integers for which $n_1 < n_2 < \cdots < n_k < \cdots$. The result we have in mind is called *Müntz's theorem*, and asserts that all real linear combinations of these functions are dense in $\mathcal{C}[0,1] \Leftrightarrow$ the series $\sum_{k=1}^{\infty} 1/n_k$ diverges. For a proof, we refer the interested reader to Lorentz [29, pp. 46–48] or Achieser [1, pp. 43–46].

Problems

1. Prove that $\mathcal{C}[a,b]$ is separable.
2. Let f be a continuous real function defined on $[0,1]$. The *moments* of f are the numbers $\int_0^1 f(x)x^n dx$, where $n = 0, 1, 2, \dots$. Prove that two continuous real functions defined on $[0,1]$ are identical if they have the same sequence of moments.
3. Use the Weierstrass theorem to prove that the polynomials are dense in $\mathcal{C}(X, \mathcal{R})$ for any closed and bounded subspace X of the real line.

36. THE STONE-WEIERSTRASS THEOREMS

Our purpose in this section is to lay bare the true nature of the Weierstrass approximation theorem. We achieve this end by generalizing the theorem in such a manner that its inessential features are stripped away.

Our starting point is the fact that the polynomials are dense in $\mathcal{C}[a,b]$ for any closed interval $[a,b]$. We wish to replace $[a,b]$ by an arbitrary compact Hausdorff space X and to make a similar statement about $\mathcal{C}(X, \mathcal{R})$. The most obvious difficulty in this program is that it is meaningless to speak of polynomials on X . This obstacle will disappear when we take a closer look at what polynomials are.

Consider the two functions 1 and x defined on $[a,b]$. The set P of all polynomials on $[a,b]$ is identical with the set of all functions which can be built from these two by applying the following three operations: multiplication, multiplication by real numbers, and addition. P is an algebra of real functions on $[a,b]$, for it is closed with respect to these three operations. Even more, it is a subalgebra of $\mathcal{C}[a,b]$. We say that P is *the subalgebra of $\mathcal{C}[a,b]$ generated by $\{1, x\}$* , for it is a subalgebra containing $\{1, x\}$ which is contained in every subalgebra with this property. We know by Problem 20-3 that the closure of a subalgebra of $\mathcal{C}(X, \mathcal{R})$ —for any topological space X —is also a subalgebra of $\mathcal{C}(X, \mathcal{R})$. We may therefore speak of the closure $\bar{P}$ of P as *the closed subalgebra of $\mathcal{C}[a,b]$ generated by $\{1, x\}$* . As above, this means that $\bar{P}$ is a closed subalgebra containing $\{1, x\}$ which is contained in every closed subalgebra

with this property. These ideas make it possible for us to state the Weierstrass theorem in the following equivalent forms:

- (1) the closed subalgebra of $\mathcal{C}[a,b]$ generated by $\{1,x\}$ equals $\mathcal{C}[a,b]$;
- (2) any closed subalgebra of $\mathcal{C}[a,b]$ which contains $\{1,x\}$ equals $\mathcal{C}[a,b]$.

These are potent statements, saying, as they do, that the very small set of functions $\{1,x\}$ suffices to generate the much more extensive set $\mathcal{C}[a,b]$. As our theorems below will show, these statements depend only on the fact that a closed subalgebra of $\mathcal{C}[a,b]$ which contains the set $\{1,x\}$ separates points in the sense of Sec. 27 (for it contains the function x) and contains all constant functions (for it contains the non-zero constant function 1).

Before we go further, it is worth observing that statement (1) is not true in general if either 1 or x is omitted from the generating set. If x is omitted, then the closed subalgebra generated by $\{1\}$ consists of the constant functions, and this is not equal to $\mathcal{C}[a,b]$ unless $a = b$. On the other hand, if 1 is omitted, then the closed subalgebra generated by $\{x\}$ contains only functions which vanish at 0, and if 0 is in $[a,b]$, then the non-zero constant functions, among others, are not in this closed subalgebra.

Our theorems rest on two lemmas, both of which have to do with the fact that $\mathcal{C}(X,R)$ is a lattice for any topological space X . If f and g are functions in $\mathcal{C}(X,R)$, we recall that their join and meet are defined by

$$(f \vee g)(x) = \max \{f(x), g(x)\}$$

and
$$(f \wedge g)(x) = \min \{f(x), g(x)\}.$$

Our first lemma states conditions which guarantee that a closed sublattice of $\mathcal{C}(X,R)$ equals $\mathcal{C}(X,R)$.

Lemma. *Let X be a compact Hausdorff space with more than one point, and let L be a closed sublattice of $\mathcal{C}(X,R)$ with the following property: if x and y are distinct points of X and a and b any two real numbers, then there exists a function f in L such that $f(x) = a$ and $f(y) = b$. Then L equals $\mathcal{C}(X,R)$.¹*

PROOF. Let f be an arbitrary function in $\mathcal{C}(X,R)$. We must show that f is in L . Let $\epsilon > 0$ be given. Since L is closed, it suffices to construct a function g in L such that $f(z) - \epsilon < g(z) < f(z) + \epsilon$ for all points z in

¹ If X has only one point, then a single constant function constitutes a closed sublattice of $\mathcal{C}(X,R)$ with the stated property which does not equal $\mathcal{C}(X,R)$. It is therefore necessary to assume that X has more than one point. Further, the reader will notice that the proof given below makes no use of the assumption that X is Hausdorff. However, if there exists a closed sublattice of $\mathcal{C}(X,R)$ with the stated property, then X is necessarily Hausdorff, so there is nothing to be gained by omitting this assumption.

X , for it will follow from this that $\|f - g\| < \epsilon$. We now construct such a function.

Let x be a point in X which is fixed throughout this paragraph, and let y be a point different from x . By our assumption about L , there exists a function f_y in L such that $f_y(x) = f(x)$ and $f_y(y) = f(y)$. Now consider the open set $G_y = \{z : f_y(z) < f(z) + \epsilon\}$. It is clear that both x and y belong to G_y , so the class of G_y 's for all points y different from x is an open cover of X . Since X is compact, this open cover has a finite subcover, which we denote by $\{G_1, G_2, \dots, G_n\}$. If the corresponding functions in L are denoted by $f_1, f_2, \dots, f_n$, then

$$g_x = f_1 \wedge f_2 \wedge \dots \wedge f_n$$

is evidently a function in L such that $g_x(x) = f(x)$ and $g_x(z) < f(z) + \epsilon$ for all points z in X .

We next consider the open set $H_x = \{z : g_x(z) > f(z) - \epsilon\}$. Since x belongs to H_x , the class of H_x 's for all points x in X is an open cover of X . The compactness of X implies that this open cover has a finite subcover, which we denote by $\{H_1, H_2, \dots, H_m\}$. We denote the corresponding functions in L by $g_1, g_2, \dots, g_m$, and we define g by $g = g_1 \vee g_2 \vee \dots \vee g_m$. It is clear that g is a function in L with the property that $f(z) - \epsilon < g(z) < f(z) + \epsilon$ for all points z in X , so our proof is complete.

In our next lemma we make use of the concept of the absolute value of a function. If f is a real or complex function defined on a topological space X , then the function $|f|$ —called the *absolute value* of f —is defined by $|f|(x) = |f(x)|$. If f is continuous, then $|f|$ is also continuous. We observe that the lattice operations in $\mathcal{C}(X, R)$ are expressible in terms of addition, scalar multiplication, and the formation of absolute values:

$$f \vee g = \frac{f + g + |f - g|}{2}$$

and

$$f \wedge g = \frac{f + g - |f - g|}{2}.$$

These identities show that any linear subspace of $\mathcal{C}(X, R)$ which contains the absolute value of each of its functions is a sublattice of $\mathcal{C}(X, R)$.

Lemma. *Let X be an arbitrary topological space. Then every closed subalgebra of $\mathcal{C}(X, R)$ is also a closed sublattice of $\mathcal{C}(X, R)$.*

PROOF. Let A be a closed subalgebra of $\mathcal{C}(X, R)$. By the above remarks, it suffices to show that if f is in A , then $|f|$ is also in A . Let $\epsilon > 0$ be given. Since $|t|$ is a continuous function of the real variable t , by the Weierstrass approximation theorem there exists a polynomial p' with the property that $||t| - p'(t)| < \epsilon/2$ for every t on the closed interval

$[-\|f\|, \|f\|]$. If p is the polynomial which results when the constant term of p' is replaced by 0, then p is a polynomial with 0 as its constant term which has the property that $||t| - p(t)| < \epsilon$ for every t in $[-\|f\|, \|f\|]$. Since A is an algebra, the function $p(f)$ in $\mathcal{C}(X, R)$ is in A . By the stated property of p , it is easy to see that $||f(x)| - p(f(x))| < \epsilon$ for every point x in X , and from this it follows that $\| |f| - p(f) \| < \epsilon$. We conclude the proof by remarking that since A is closed, the fact that $|f|$ can be approximated by the function $p(f)$ in A shows that $|f|$ is in A .

We are now in a position to prove the Stone-Weierstrass theorems.

Theorem A (the Real Stone-Weierstrass Theorem). *Let X be a compact Hausdorff space, and let A be a closed subalgebra of $\mathcal{C}(X, R)$ which separates points and contains a non-zero constant function. Then A equals $\mathcal{C}(X, R)$.*

PROOF. If X has only one point, then $\mathcal{C}(X, R)$ contains only constant functions; and since A contains a non-zero constant function and is an algebra, it contains all constant functions and equals $\mathcal{C}(X, R)$. We may thus assume that X has more than one point. By the above lemmas, it suffices to show that if x and y are distinct points of X , and if a and b are any two real numbers, then there exists a function f in A such that $f(x) = a$ and $f(y) = b$. Since A separates points, there exists a function g in A such that $g(x) \neq g(y)$. If we now define f by

$$f(z) = a \frac{g(z) - g(y)}{g(x) - g(y)} + b \frac{g(z) - g(x)}{g(y) - g(x)},$$

then f clearly has the required properties.

We next turn our attention to the complex case, that is, to conditions which guarantee that a closed subalgebra of $\mathcal{C}(X, C)$ equals $\mathcal{C}(X, C)$. It is first of all necessary to understand that the conditions of Theorem A will not suffice. The simplest example which shows this requires a little knowledge of the theory of analytic functions, and the reader without such knowledge may skip at once to the next paragraph. Let X be the closed unit disc $\{z: |z| \leq 1\}$ in the complex plane. X is clearly a compact Hausdorff space. Consider the set A of all functions in $\mathcal{C}(X, C)$ which are analytic in the interior of X . A is evidently a subalgebra of $\mathcal{C}(X, C)$, and one sees that it is closed by using Morera's theorem. A separates points, for it contains the function f defined by $f(z) = z$. It also contains all constant functions. In spite of this, A does not equal $\mathcal{C}(X, C)$; for the function g defined by $g(z) = \bar{z}$ is in $\mathcal{C}(X, C)$ but is not in A , since it is not differentiable at any point.

What can be done to salvage Theorem A in the complex case? The answer lies in the operation of conjugation discussed at the end of Sec. 20. If f is a complex function defined on a topological space X , we

remind the reader that its *conjugate* $\bar{f}$ is defined by $\bar{f}(x) = \overline{f(x)}$. It will also be convenient for us to define the *real part* and the *imaginary part* of f :

$$R(f) = \frac{f + \bar{f}}{2} \quad \text{and} \quad I(f) = \frac{f - \bar{f}}{2i}. \quad (1)$$

We observe that if a complex function f has different values at two distinct points of X , then at least one of the functions $R(f)$ and $I(f)$ also has different values at these points.

Theorem B (the Complex Stone-Weierstrass Theorem). *Let X be a compact Hausdorff space, and let A be a closed subalgebra of $\mathcal{C}(X, \mathbb{C})$ which separates points, contains a non-zero constant function, and contains the conjugate of each of its functions. Then A equals $\mathcal{C}(X, \mathbb{C})$.*

PROOF. The real functions in A clearly form a closed subalgebra B of $\mathcal{C}(X, \mathbb{R})$. Let us assume for a moment that B equals $\mathcal{C}(X, \mathbb{R})$. If f is an arbitrary function in $\mathcal{C}(X, \mathbb{C})$, then $R(f)$ and $I(f)$ are in $\mathcal{C}(X, \mathbb{R})$, and are thus in B . But since $f = R(f) + iI(f)$ and A is an algebra, f is in A and A equals $\mathcal{C}(X, \mathbb{C})$. It therefore suffices to show that B equals $\mathcal{C}(X, \mathbb{R})$. We prove this by applying Theorem A.

We begin by showing that B separates points. Let x and y be distinct points in X . Since A separates points, there exists a function f in A which has different values at x and y . As we saw in the above remarks, $R(f)$ or $I(f)$ also has different values at x and y . Since A is an algebra which contains the conjugate of each of its functions, formulas (1) show that both $R(f)$ and $I(f)$ are in B , so B separates points. We next show that B contains a non-zero constant function. By our hypothesis, A contains some non-zero constant function g . A is an algebra which contains the conjugate of each of its functions, so $g\bar{g} = |g|^2$ is a non-zero constant function in B . Theorem A now implies directly that B equals $\mathcal{C}(X, \mathbb{R})$, and our proof is complete.

The two Stone-Weierstrass theorems are among the most important facts in modern analysis. The theory developed in the last three chapters of this book could hardly exist without them, and they have many other applications as well.¹

Problems

1. Prove the two-variable Weierstrass approximation theorem: if $f(x, y)$ is a real function defined and continuous on the closed rectangle $X = [a, b] \times [c, d]$ in the Euclidean plane $\mathbb{R}^2$, then f can be uniformly approximated on X by polynomials in x and y with real coefficients.

¹ See Stone [40].

2. Let X be the closed unit disc in the complex plane, and show that any function f in $\mathcal{C}(X, \mathbb{C})$ can be uniformly approximated on X by polynomials in z and $\bar{z}$ with complex coefficients.
3. Let X and Y be compact Hausdorff spaces, and f a function in $\mathcal{C}(X \times Y, \mathbb{C})$. Show that f can be uniformly approximated by functions of the form $\sum_{i=1}^n f_i g_i$, where the f_i 's are in $\mathcal{C}(X, \mathbb{C})$ and the g_i 's are in $\mathcal{C}(Y, \mathbb{C})$.

37. LOCALLY COMPACT HAUSDORFF SPACES

In Sec. 23 we defined a *locally compact space* to be a topological space in which each point has a neighborhood with compact closure. Locally compact spaces often arise in the applications of topology to geometry and analysis, and since those which do are almost always Hausdorff spaces, we restrict our attention in this section to locally compact Hausdorff spaces.

The main fact about such a space is that it can be converted into a compact Hausdorff space by suitably adjoining a single point. The reader is perhaps familiar from analysis with the prototype of this process, in which the complex plane $\mathbb{C}$ is enlarged by adjoining to it an "ideal point" called the *point at infinity* and denoted by ∞ . This ideal point can be thought of as any object not in $\mathbb{C}$, and we denote by $\mathbb{C}_\infty$ the larger set $\mathbb{C} \cup \{\infty\}$. $\mathbb{C}_\infty$ is called the *extended complex plane* when the neighborhoods of ∞ (other than $\mathbb{C}_\infty$ itself) are taken to be the complements in $\mathbb{C}_\infty$ of the closed and bounded subsets (i.e., the compact subspaces) of $\mathbb{C}$.

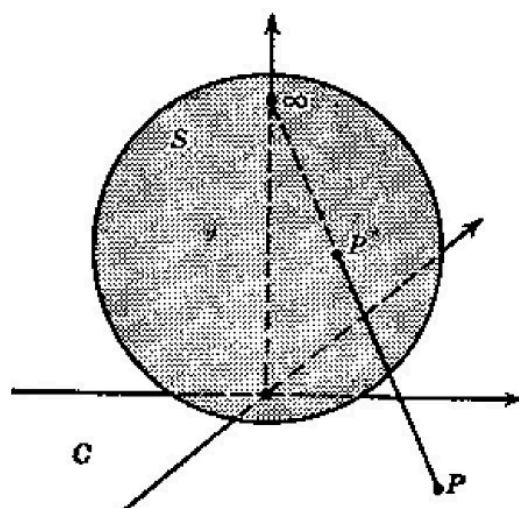


Fig. 27. The Riemann sphere.

These ideas add nothing to our understanding of the complex plane, but they do clarify many proofs and simplify the statements of many theorems, and they are valuable for this reason. Figure 27 gives an easy way of visualizing the extended complex plane. In this figure, the surface S of a sphere of radius $\frac{1}{2}$ is rested tangentially on $\mathbb{C}$ at the origin. It is

customary to call the point of contact the *south pole* and the opposite point the *north pole*. The indicated projection from the north pole establishes a homeomorphism between S minus its north pole and $\mathbb{C}$, so